EFC Phase 3: SPARC Validation

Empirical Test of Energy-Flow Cosmology
Against the Radial Acceleration Relation

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Abstract

We perform an unbinned maximum-likelihood fit with intrinsic scatter of the EFC screening form $\ln\mu = k\ln(1+g^\dagger/g_{\rm bar})$ to 174 galaxies from the SPARC database (3264 data points). We obtain $k = 0.415 \pm 0.029$ and $g^\dagger = (2.51 \pm 0.60) \times 10^{-10}$ m/s^2, with $\sigma_{\rm int} = 0.33$ in $\ln\mu$ (≈ 0.14 dex). Using the independently obtained coupling $a_G = 0.094 \pm 0.01$ from the H_0 regime analysis [2], we infer the phase-gain parameter $C = k/a_G = 4.4 \pm 0.6$. This provides a quantitative cross-scale consistency check within the EFC projection framework. A parameter-free prediction requires an independent derivation of $g^\dagger$ and C.

Keywords: radial acceleration relation, MOND, modified gravity, Hubble tension, SPARC

1 Introduction

The radial acceleration relation (RAR) demonstrates a tight empirical correlation between observed centripetal acceleration $g_{\rm obs}$ and baryonic acceleration $g_{\rm bar}$ in rotationally-supported galaxies [5]. This correlation, with intrinsic scatter of ∼0.13 dex, poses a challenge for ΛCDM while being naturally explained by modified gravity theories.

In the EFC framework, gravitational enhancement arises from entropy-phase accumulation $\Delta\Phi$ along the entropy axis S. The Core Lock mechanism [1] specifies $G_{\rm eff} = G_0 \exp(a_G \cdot \Delta\Phi)$, where $a_G \approx 0.094$ is derived from H_0 tension analysis [2].

A screening ansatz—where phase accumulation is suppressed when baryonic gravity dominates—yields $\Delta\Phi = C \cdot \ln(1 + g^\dagger/g_{\rm bar})$. The gravitational enhancement becomes:

$$\mu = \left(1 + \frac{g^\dagger}{g_{\rm bar}}\right)^k, \quad k = a_G \cdot C \tag{1}$$

2 Data and Method

We use the SPARC database [4] containing 175 disk galaxies with Spitzer 3.6μm photometry and high-quality HI/Hα rotation curves. After quality cuts ($Q \leq 2$, inclination > 30), 174 galaxies with 3264 data points remain.

For each point, $g_{\rm bar}$ is computed from the baryonic mass model and $g_{\rm obs}$ from the observed rotation velocity. We define $y_i = \ln\mu_i = \ln(g_{{\rm obs},i}/g_{{\rm bar},i})$ and fit to the model $y_{\rm model} = k\ln(1 + g^\dagger/g_{\rm bar})$.

2.1 Likelihood

We use maximum likelihood with intrinsic scatter σ_{int}:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_i \left[\frac{(y_i - k \ln(1 + g^\dagger / g_{\text{bar},i}))^2}{\sigma_{y,i}^2 + \sigma_{\text{int}}^2} + \ln(\sigma_{y,i}^2 + \sigma_{\text{int}}^2) \right] \tag{2}$$

where $\sigma_{y,i}$ is the measurement uncertainty on $\ln \mu_i$, propagated from the SPARC-provided uncertainties on V_{obs} and the mass model via $\sigma_y \approx 2\sigma_V / V$.

Bootstrap resampling (500 iterations) provides uncertainty estimates. We resample galaxies with replacement and include all radial points per selected galaxy, preserving intra-galaxy correlations.

3 Results

3.1 Canonical Fit

Parameter	Value	Source
k (screening exponent)	0.415 ± 0.029	SPARC fit (bootstrap)
$g^\dagger$ (transition scale)	$(2.51 \pm 0.60) \times 10^{-10}$ m/s^2	SPARC fit
σ_{int} (intrinsic scatter)	0.33 in $\ln \mu$ (≈ 0.14 dex)	SPARC fit
a_G (universal coupling)	0.094 ± 0.01	H_0 analysis [2]
$C = k/a_G$ (phase-gain)	4.4 ± 0.6	Derived

Table 1: Summary of fitted and derived parameters. The intrinsic scatter $\sigma_{\text{int}} = 0.33$ in $\ln \mu$ corresponds to $0.33/\ln 10 \approx 0.14$ dex, consistent with RAR literature values. Uncertainty on C is propagated from k and a_G.

3.2 Robustness

Alternative fitting approaches (binned data, different quality cuts, per-galaxy median) yield k in the range 0.38–0.45, with $g^\dagger$ in the range $(2.0\text{–}3.0) \times 10^{-10}$ m/s^2. The canonical values in Table 1 represent the global unbinned fit with intrinsic scatter.

3.3 Comparison with MOND

The fitted $g^\dagger = 2.5 \times 10^{-10}$ m/s^2 differs from the canonical MOND scale $a_0 = 1.2 \times 10^{-10}$ m/s^2 due to the different functional forms: MOND uses $\mu = x/(1+x)$ with $x = g/a_0$, while EFC uses $\mu = (1 + g^\dagger/g)^k$.

4 Discussion

The parameter $C = k/a_G$ connects galactic and cosmological scales. Given that a_G is independently derived from H_0 tension analysis (not from RAR), the inferred $C \approx 4.4$ serves as a cross-scale consistency check.

Physically, C represents the ratio of galactic to cosmological phase accumulation. In the H_0 analysis, $\Delta\Phi_{\text{cosmo}} \approx 1.7$ spans the background-to-structure transition. The galactic phase $\Delta\Phi_{\text{gal}} = C \cdot \ln(1 + g^\dagger/g_{\text{bar}})$ varies with local acceleration: at transition ($g_{\text{bar}} = g^\dagger$), $\Delta\Phi_{\text{gal}} \approx 3.1$; in outer disks ($g_{\text{bar}} = 0.1 g^\dagger$), $\Delta\Phi_{\text{gal}} \approx 10.6$. This implies a regime mismatch factor $\Delta\Phi_{\text{gal}}/\Delta\Phi_{\text{cosmo}} \sim 2\text{–}6$ for typical transition-to-outer-disk values.

A full parameter-free prediction would require deriving both $g^\dagger$ and C from EFC first principles, rather than fitting them to data.

5 Conclusion

EFC's screening form provides a good empirical description of SPARC RAR, with $k = 0.415 \pm 0.029$ and intrinsic scatter $\sigma_{\text{int}} \approx 0.14$ dex. Combined with the independently derived $a_G = 0.094$ from H_0 analysis, this yields $C = k/a_G = 4.4 \pm 0.6$ as a cross-scale consistency parameter.

This work establishes RAR compatibility; a stronger claim requires independent derivation of $g^\dagger$ and C from the Core Lock framework.

References

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